

Dopants at the Atomic Limit in Silicon and Germanium

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Professor of Physics

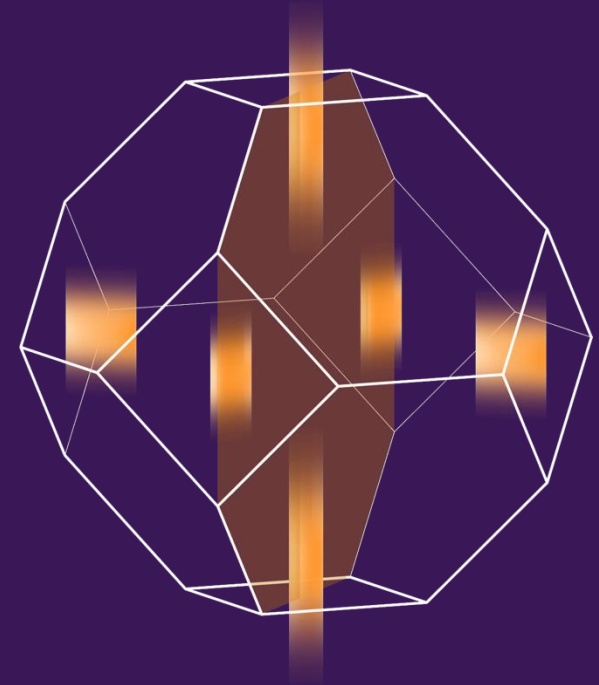
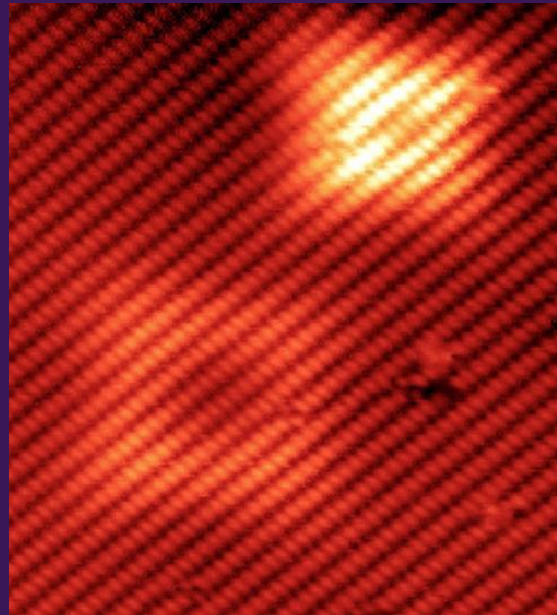
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Engineering in Semiconductor Technology*

Plenary Lecture

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Flatté

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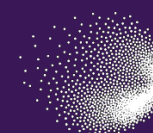
Vladimir
Strocov

Yasin
Ekinci



Gabriel
Aeppli

Funding and Institutions



PSI

EPSRC

Pioneering research
and skills



Current group in the LT-STM lab at UCL

Single donors placed with atomic precision are a route to scalable qubits

- **Donor spins are near-ideal qubits**

Identical; seconds-scale coherence in $^{28}\text{Si}^1$

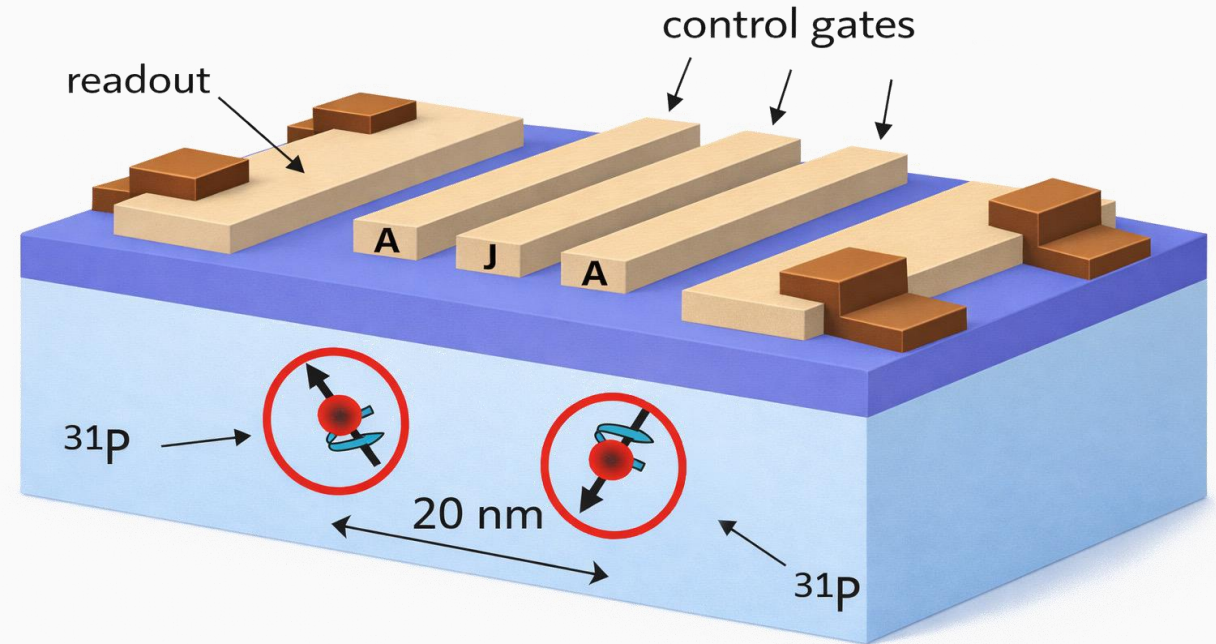
- **Coupling needs atomic placement**

Useful exchange coupling requires closely spaced donors (~ 20 nm), registered to gates; valley interference makes coupling sensitive to single-site shifts²

- **STM lithography enables dopant placement**

Deterministic single-atom placement

→ can we achieve scalable donor arrays?



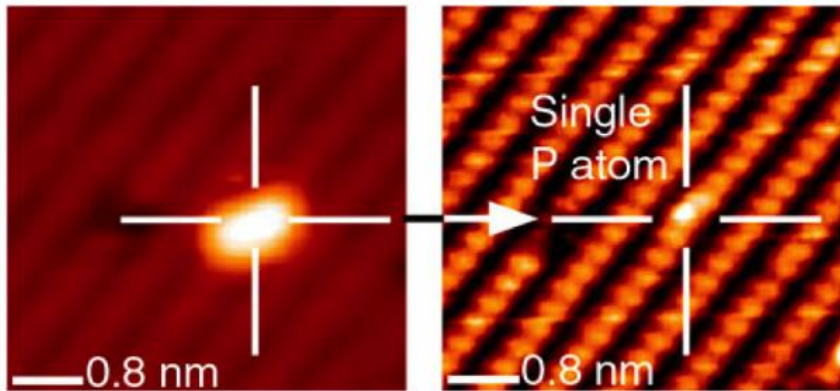
Kane, Nature 393 133 (1998)

¹ Muhonen et al., *Nat. Nanotechnol.* 9, 986 (2014)

² Koiller, Hu & Das Sarma, *Phys. Rev. Lett.* 88, 027903 (2002)

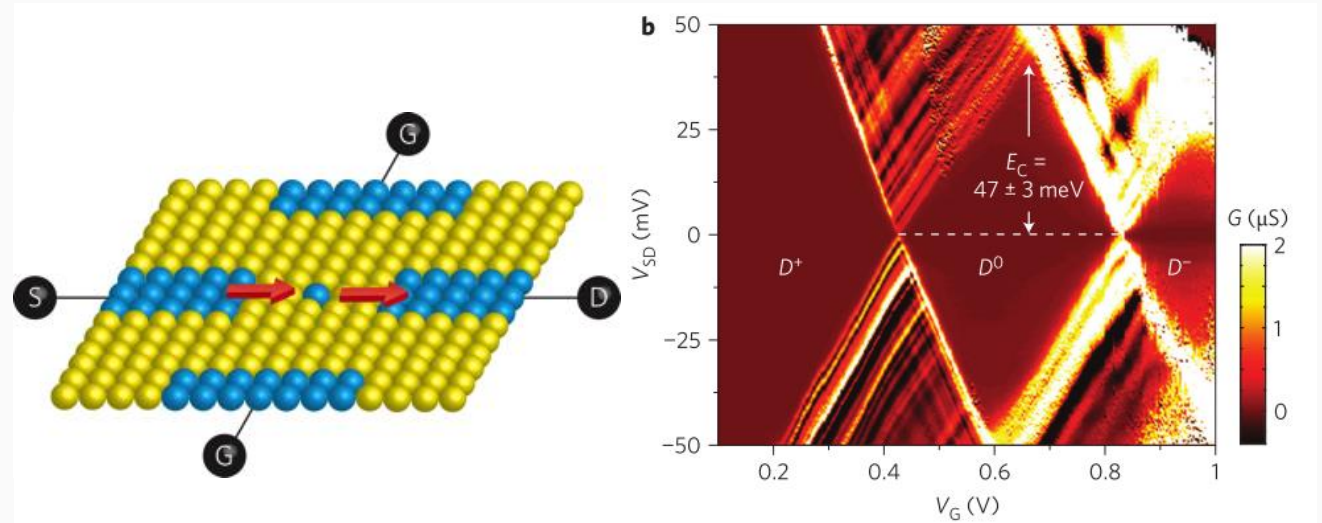
Single P atoms placed within 1 nm², enabling a single-atom transistor

Single dopants positioned (2003)



Schofield, et al., PRL 91, 136104 (2003)

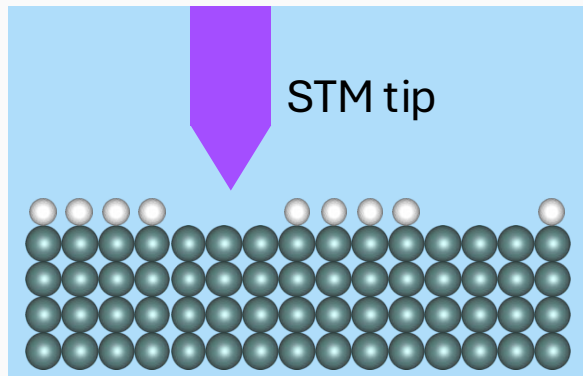
Single Atom Transistor (2012)



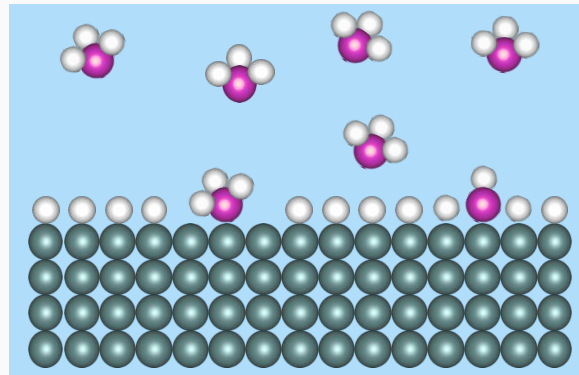
Fueschle et al., Nature Nano. 7, 242 (2012)

STM hydrogen lithography defines where each dopant incorporates

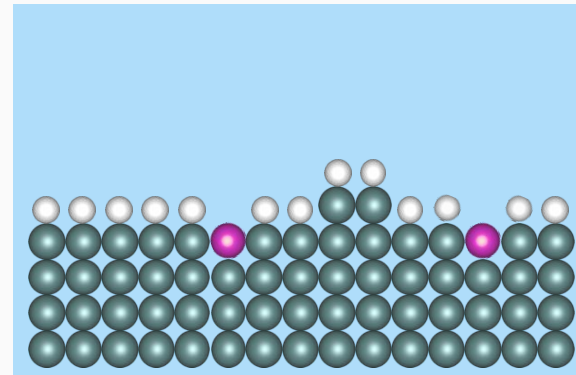
1. STM patterning of Si(001):H



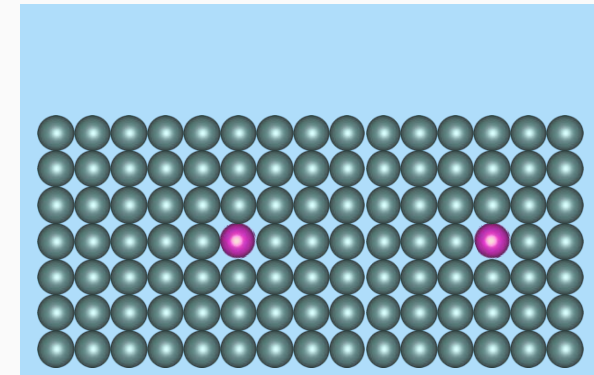
2. Precursor (PH_3) adsorption on exposed Si



3. Anneal $\rightarrow$ dopant incorporation (350°C)



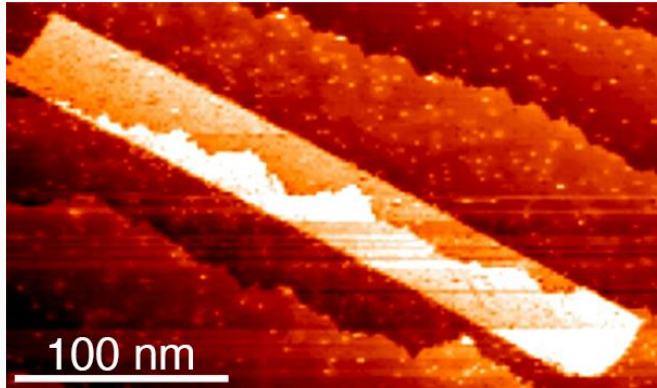
4. Epitaxial overgrowth (encapsulation)



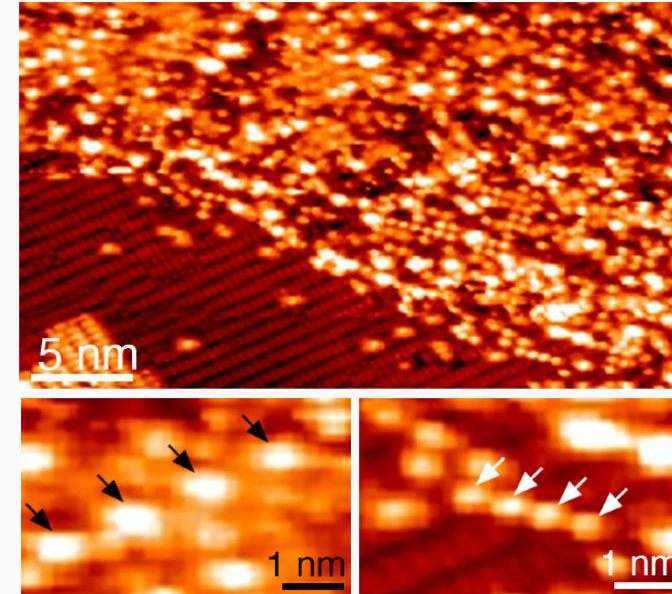
● Silicon ● Dopant ● Hydrogen

Controlled adsorption of PH_3 to $\text{Si}(001)$

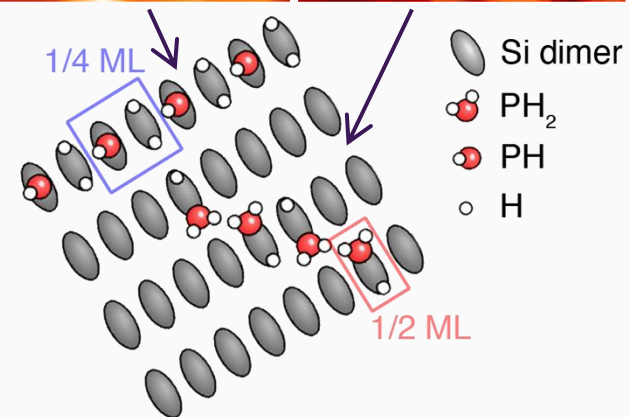
300 x 50 nm² patch of bare Si



Zoom-in after PH_3 dosing



- H-desorption → bare Si patch
- PH_3 adsorption within patterned region
- Mixed PH_x configurations within patches
- Surrounding H-terminated surface defines adsorption registry

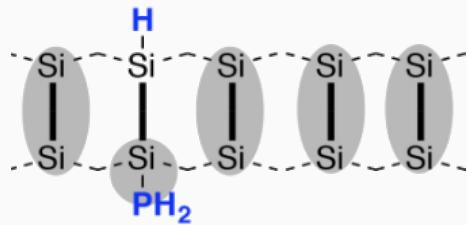
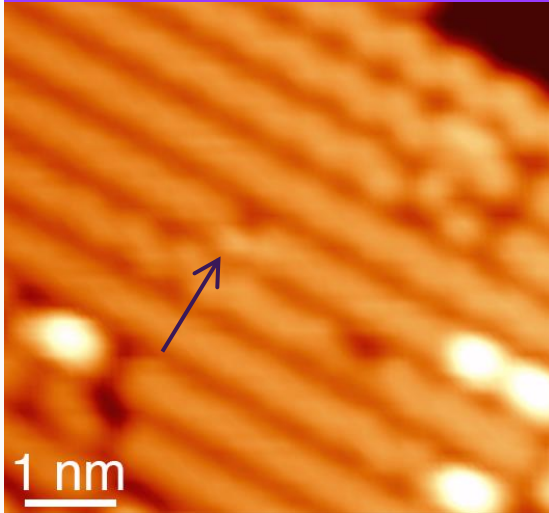


1 ML = $6.8 \times 10^{14} \text{ cm}^{-2}$
(Si(001) surface atoms)

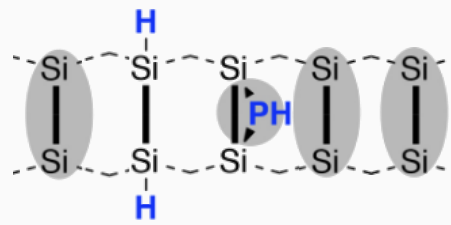
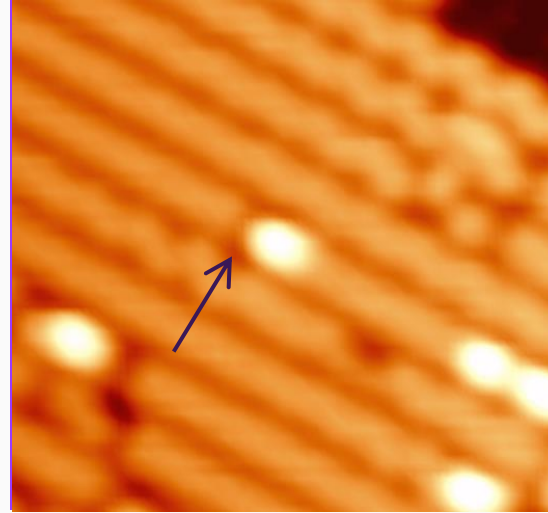
Schofield, et al., *JPCB*, 110, 3173 (2006)

Stepwise PH_3 dissociation on $\text{Si}(001)$ at RT

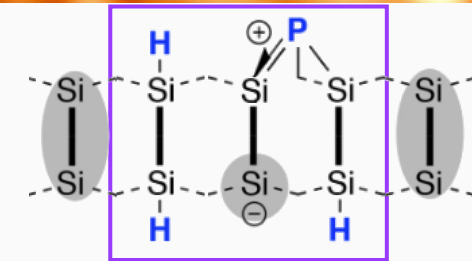
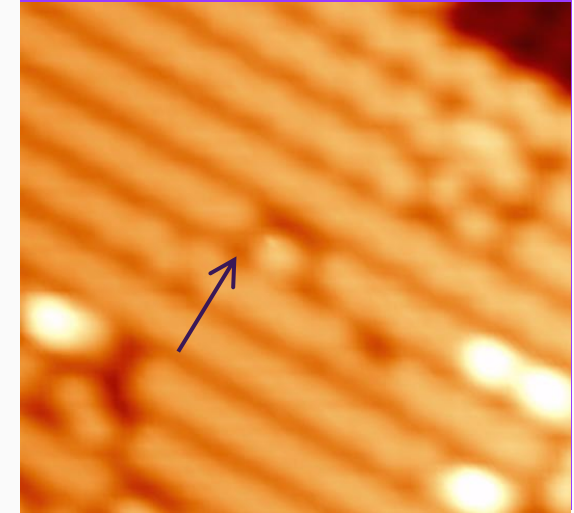
Initial adsorption: $\text{PH}_2 + \text{H}$



Partial dissociation: $\text{PH} + 2\text{H}$



Complete dissociation: $\text{P} + 3\text{H}$



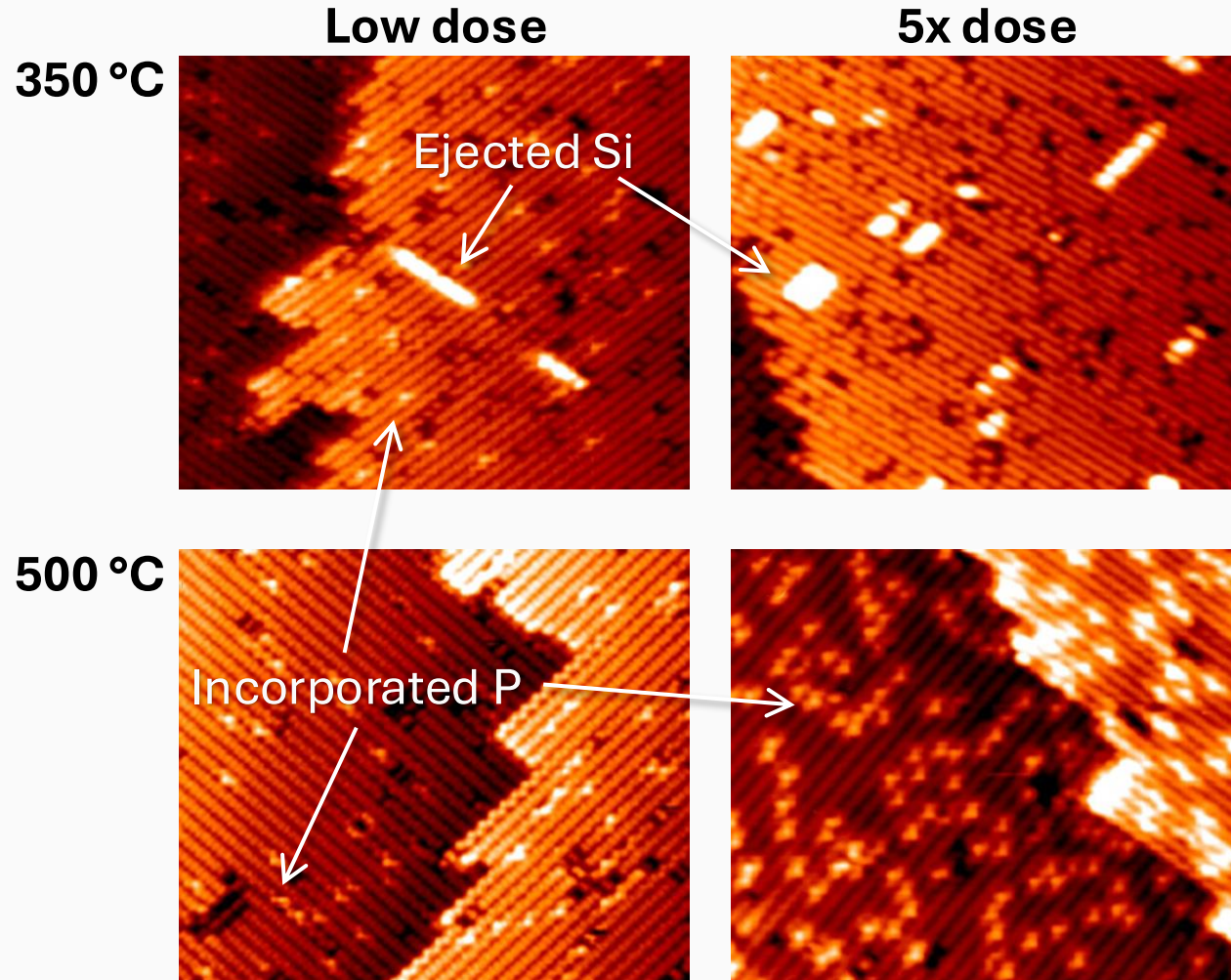
1/6 ML

- Each intermediate structure is directly resolved in STM (time scale $\sim$ hours at RT)

Annealing to incorporate P into the Si(001) surface

350 °C:

- Onset of P incorporation (asymmetric features)
- Si ejection → epitaxial chains
- Residual H

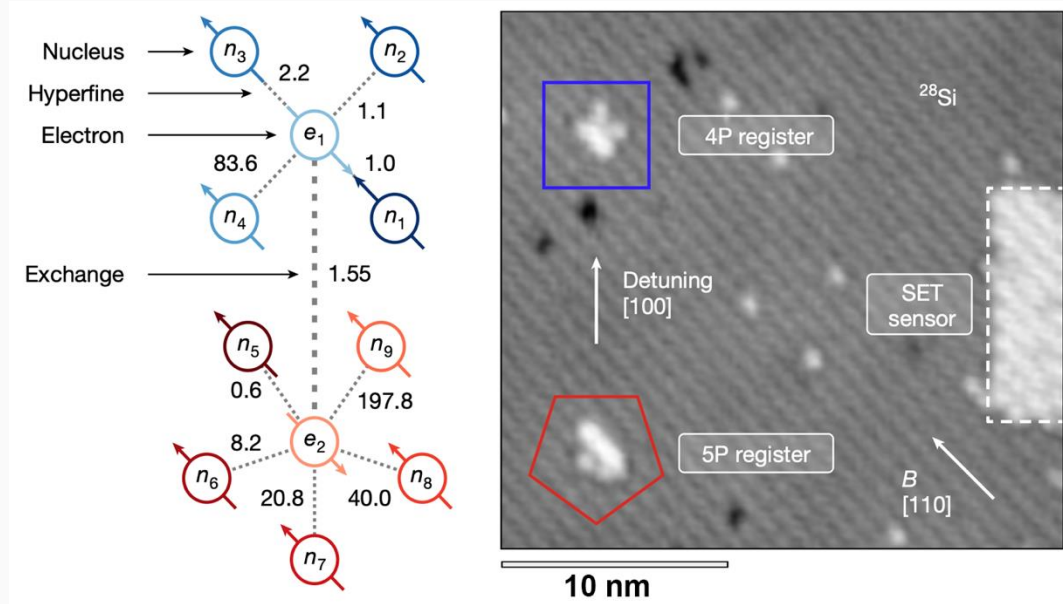


500 °C:

- Surface clean
- Only incorporated P remains

Curson et al., PRB 69, 195303 (2004)

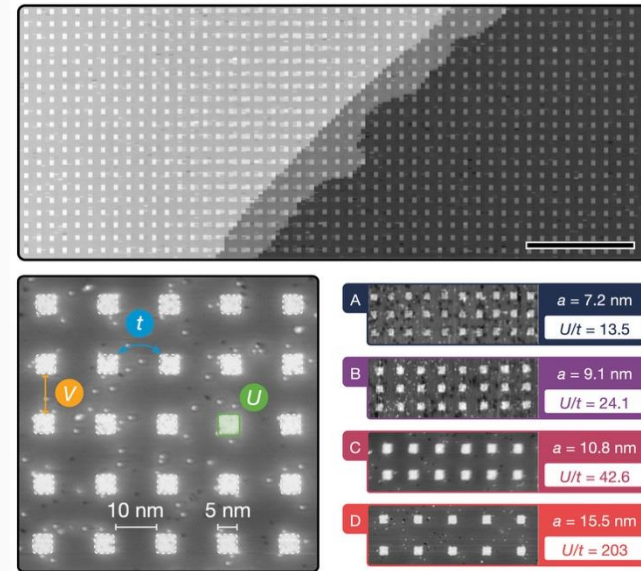
State of the art: STM-placed donor clusters now run 11-qubit processors and 15,000-dot simulators



Edlbauer *et al.*, Nature 648, 569 (2025)

Quantum processing

- 11 qubits in two exchange-coupled registers of **4–5 P each**
- Two-qubit gates at **99.9%** fidelity, above the ~99% error-correction threshold



Donnelly *et al.*, Nature 650, 574 (2026)

Quantum simulation

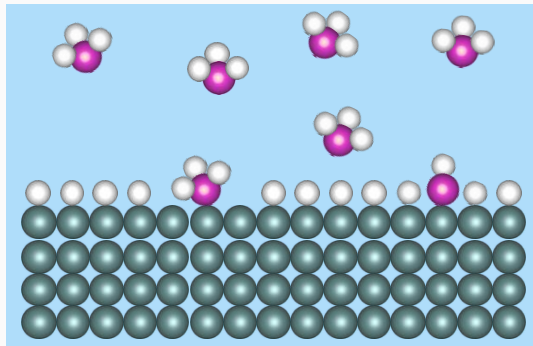
- 15,000 25 nm^2 dots of **~50 P each**, tunable t , U by spacing and size
- Proof of principle Mott–Hubbard / Anderson metal–insulator transition demonstrations

Arsine lifts donor incorporation from ~70% to ~98%

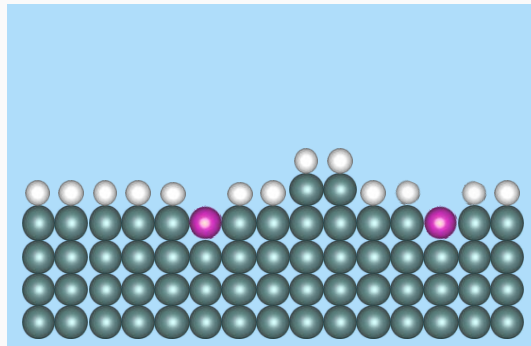
Single-dopant incorporation

P ~70% · As ~98%

2. PH_3 / AsH_3 adsorption
on exposed Si

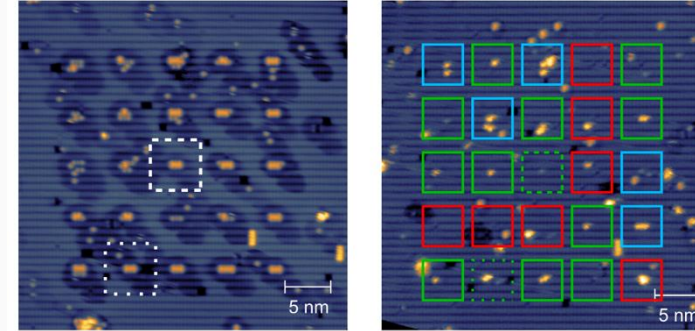


3. Anneal $\rightarrow$ dopant
incorporation (350°C)

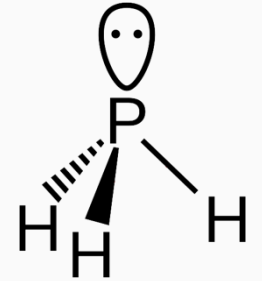


● Silicon ● Dopant ● Hydrogen

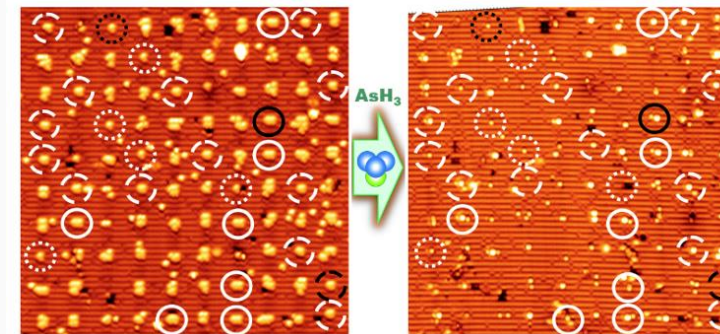
Phosphorus incorporation ~70%



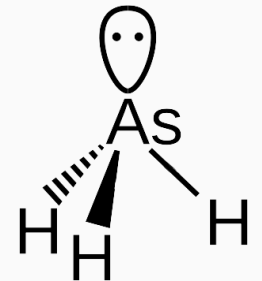
Ivie et al., Phys. Rev. Appl. 16, 054037 (2021) – Sandia group
Martin Fuchsle PhD Thesis (UNSW, 2011) – UNSW group



Arsenic incorporation ~98%



Stock, Advanced Materials, 2312282 (2024)



SX-ARPES reads buried δ -layer band structure non-destructively

Method:

- Momentum-resolved electronic structure via SX-ARPES

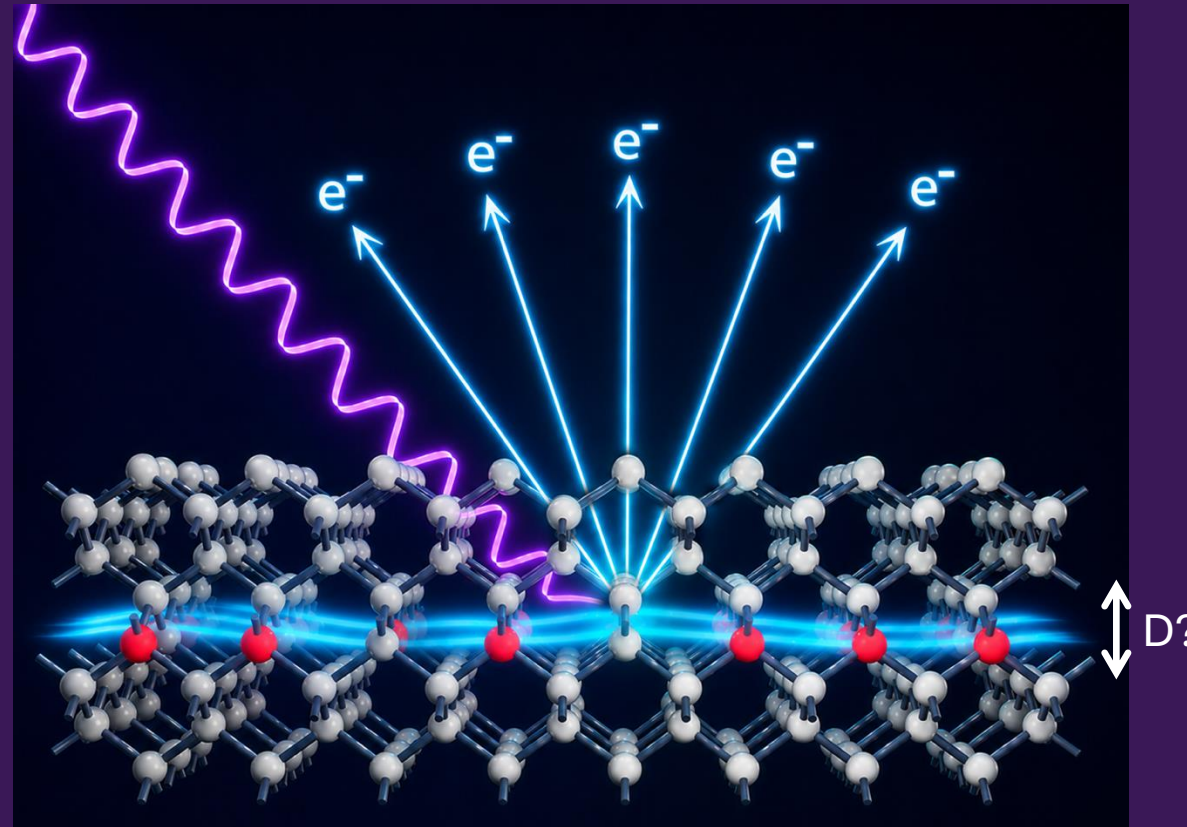
Why Soft X-rays?

- Soft x-rays (300–1500 eV)
→ IMFP $\approx$ 2 nm
→ subsurface sensitivity and reduced IMFP broadening

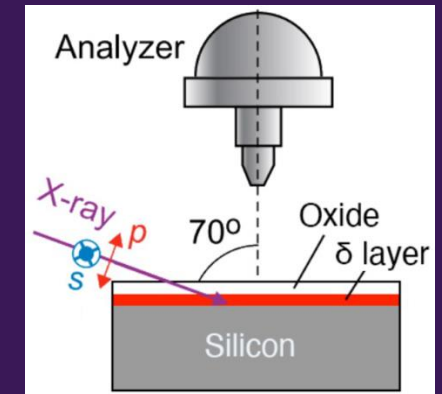
Experimental regime:

- $T = 12$ K
- $\Delta E = 50\text{--}90$ meV, $\Delta\theta = 0.1^\circ$
- Spot size $\sim 10 \times 72 \mu\text{m}^2$

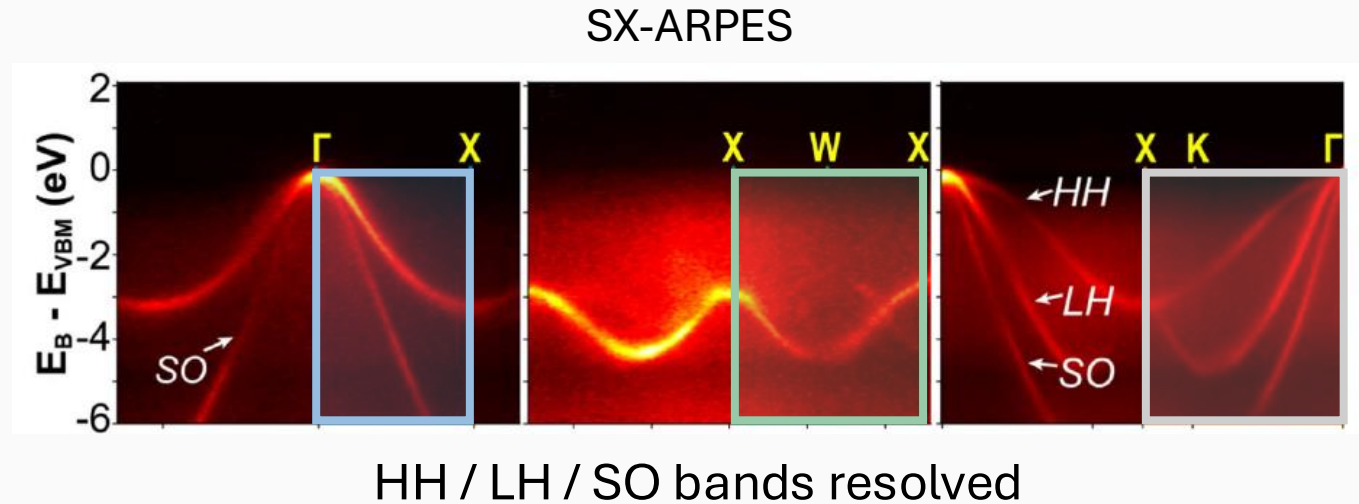
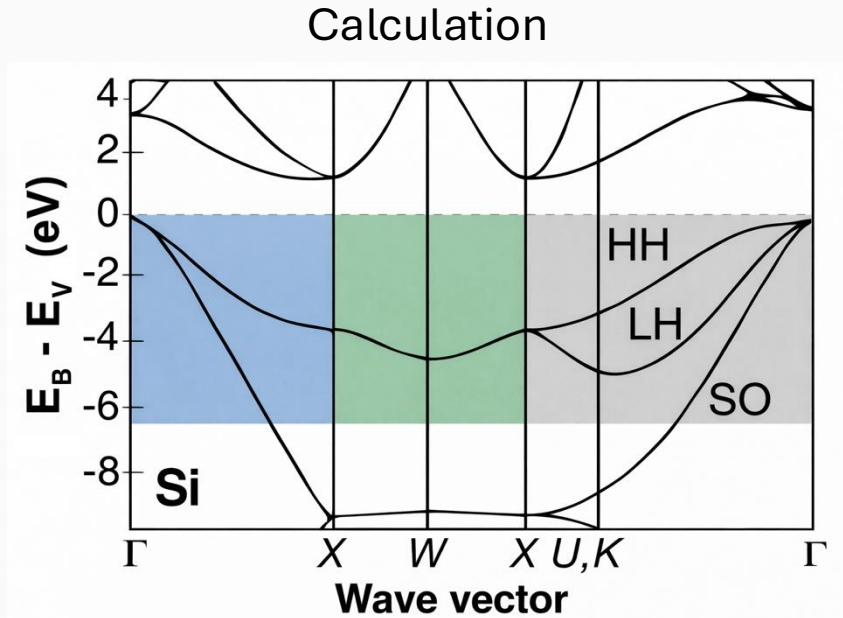
Photoemission from buried δ -layer



Measurement geometry



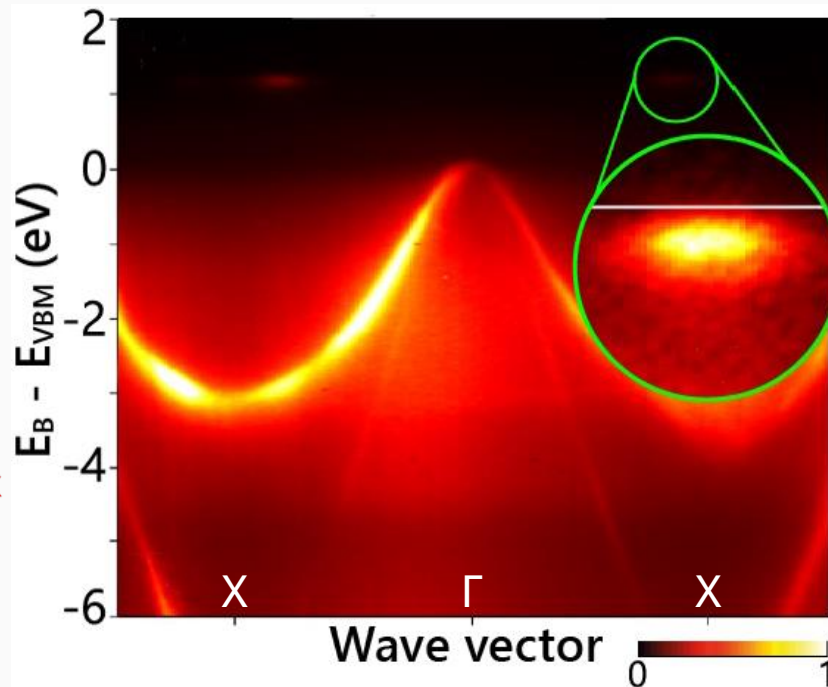
ARPES directly resolves Si valence band structure



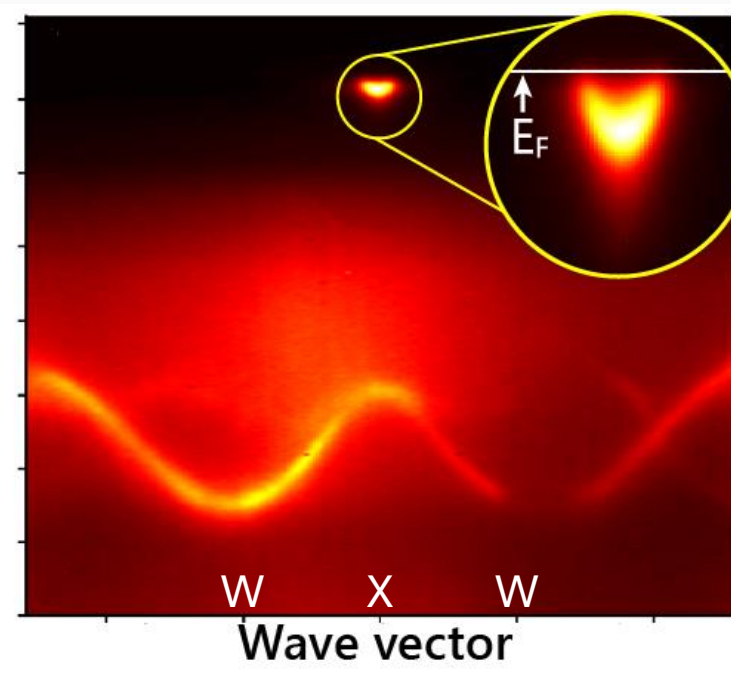
Bulk reference: validates k -space calibration before δ -layer data

Out-of-plane (Γ) valleys lie below in-plane (Δ) valleys: confinement is valley-dependent

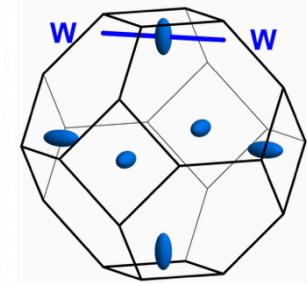
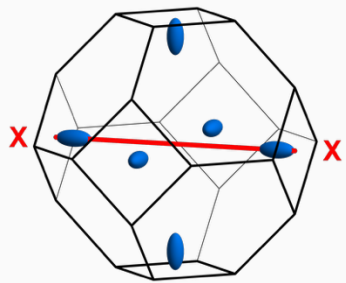
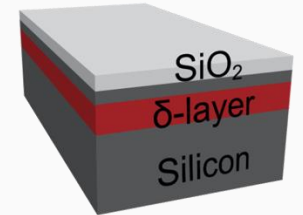
Γ -X (in-plane)



X-W (out-of-plane)

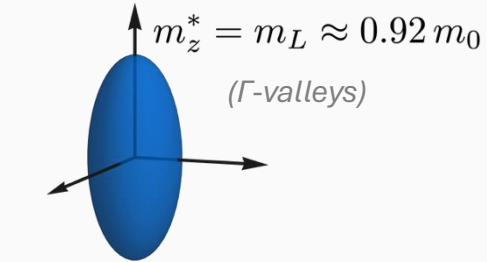


δ -layer confinement
(2 nm overgrowth)

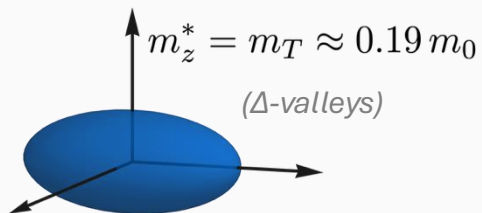
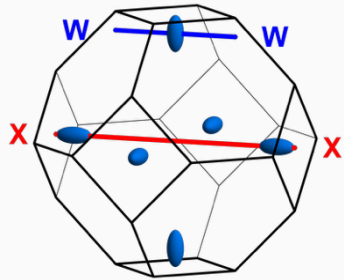


Heavy $m_z^* = m_L$ lowers Γ -valley confinement energy

Effective mass along k_z



Γ and Δ valley locations;
labels refer to the 2D projected BZ



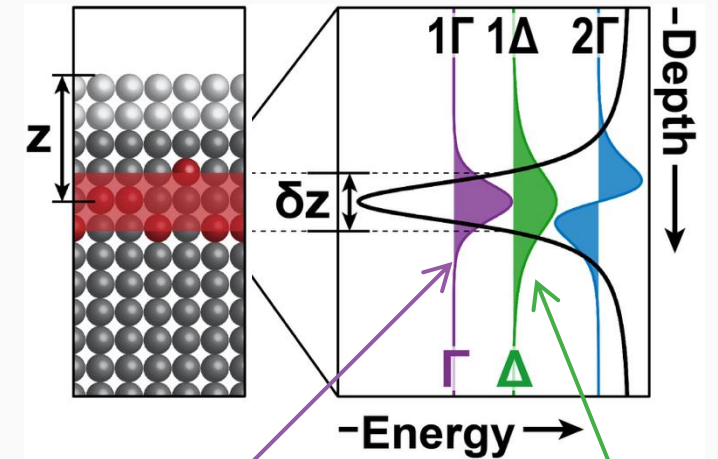
1D confinement energy

$$\left(-\frac{\hbar^2}{2m_z^*} \frac{d^2}{dz^2} + V(z) \right) \psi(z) = E \psi(z)$$

$$E_n \approx \frac{\hbar^2 \pi^2 n^2}{2m_z^* L^2} \rightarrow E_n \propto \frac{1}{m_z^*}$$

Larger $m_z^* \rightarrow$ lower zero-point KE $\rightarrow$ lower E_1

Electrostatic confinement $V(z)$

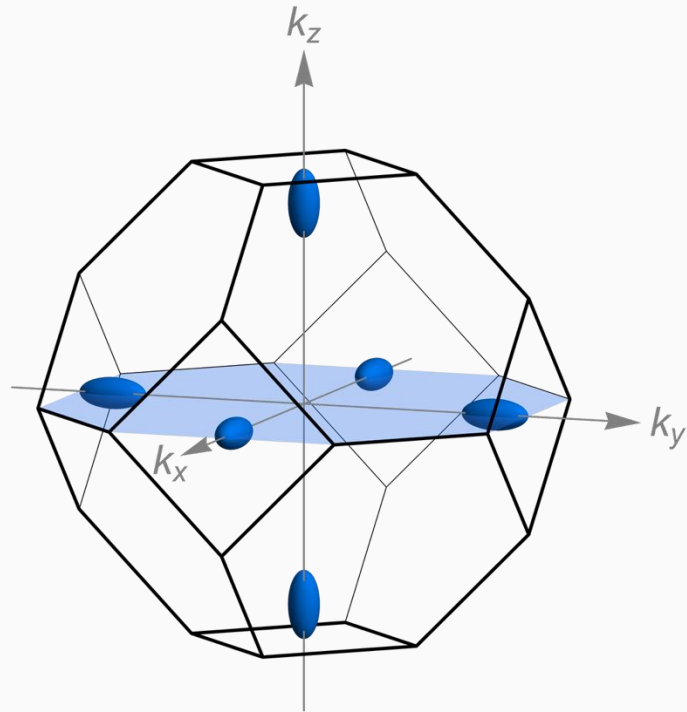


Out-of-plane E_1
 ~ 150 meV

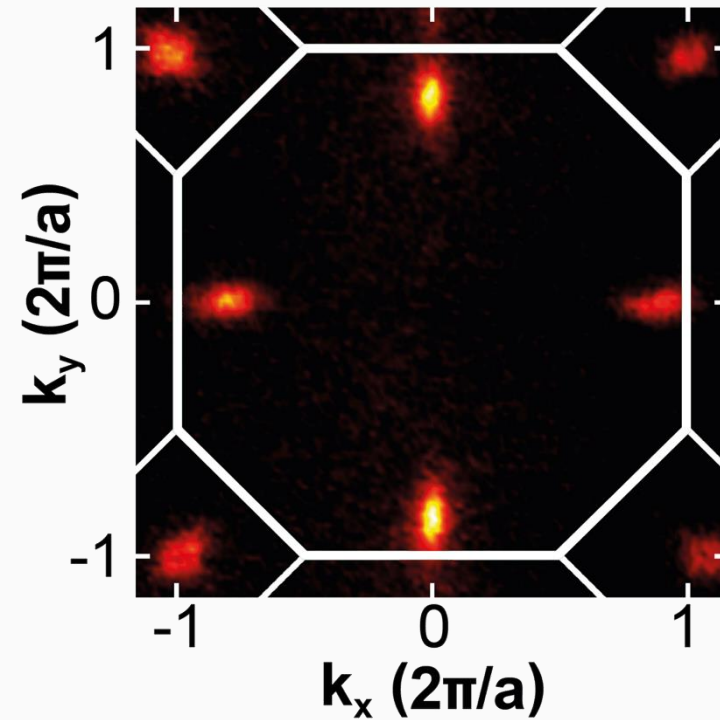
In-plane E_1
 ~ 100 meV

In-plane Fermi surface: four Δ pockets at 0.85 Γ -X, as in bulk Si

3D conduction band

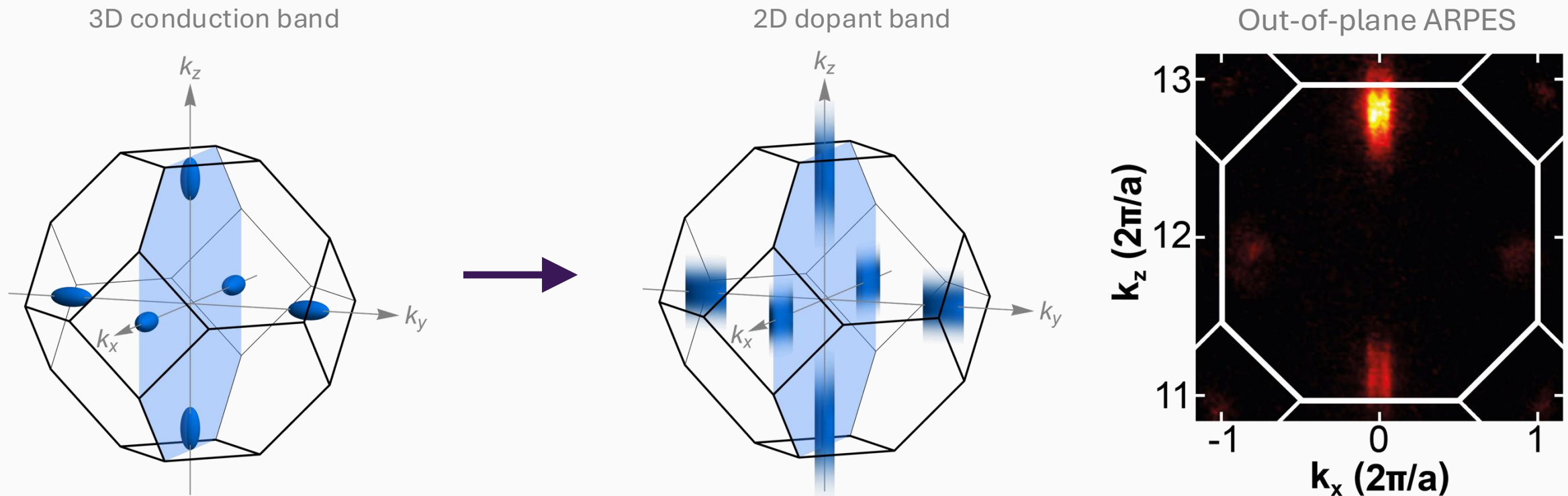


In-plane ARPES



No k_z dispersion $\rightarrow$ δ -layer electrons are 2D confined

The dispersionless, streak-like intensity (independent of k_z) is the direct signature of confinement along z .



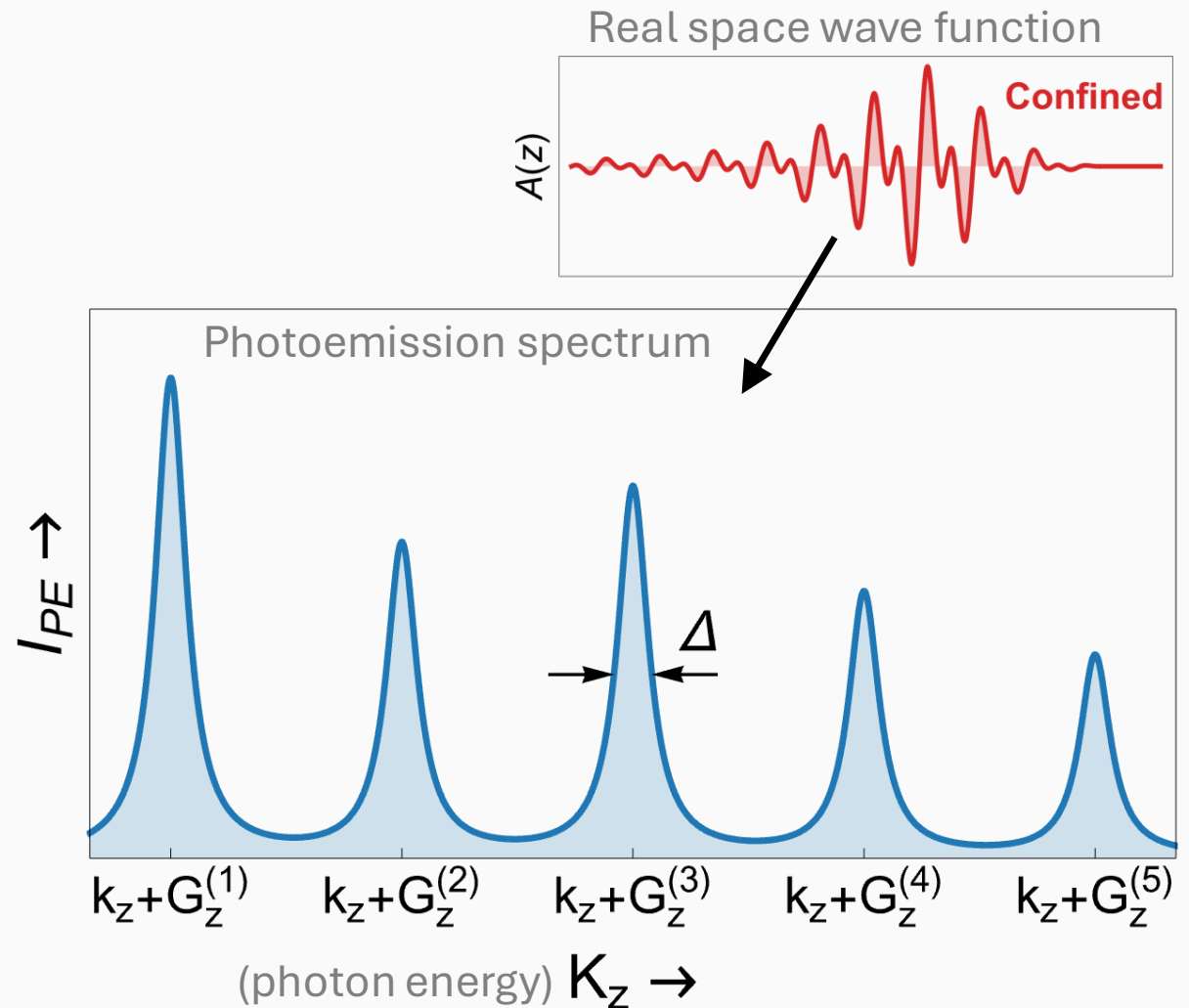
Photoemission along k_z : peaks at $k_z + G_z$, width set by confinement and IMFP

- Photoemission intensity is proportional to the Fourier transform of the out-of-plane wave function:

$$I_{\text{PE}}^{\mathbf{K}_{\parallel}} \propto \sum_{G_z} \left| C_{\mathbf{K}_{\parallel}}^{G_z} \right|^2 \left| \mathcal{F}_{K_z - k_z - G_z} \left(e^{-K_z^{\text{im}} z} \mathcal{E}(z) \right) \right|^2$$

- Peaks at $K_z = k_z + G_z^{(n)}$ weighted by $\left| C_{K_{\parallel}}^{G_z} \right|^2$

These Bloch coefficients encode the crystal band structure, and the peak heights map the Bloch wave composition

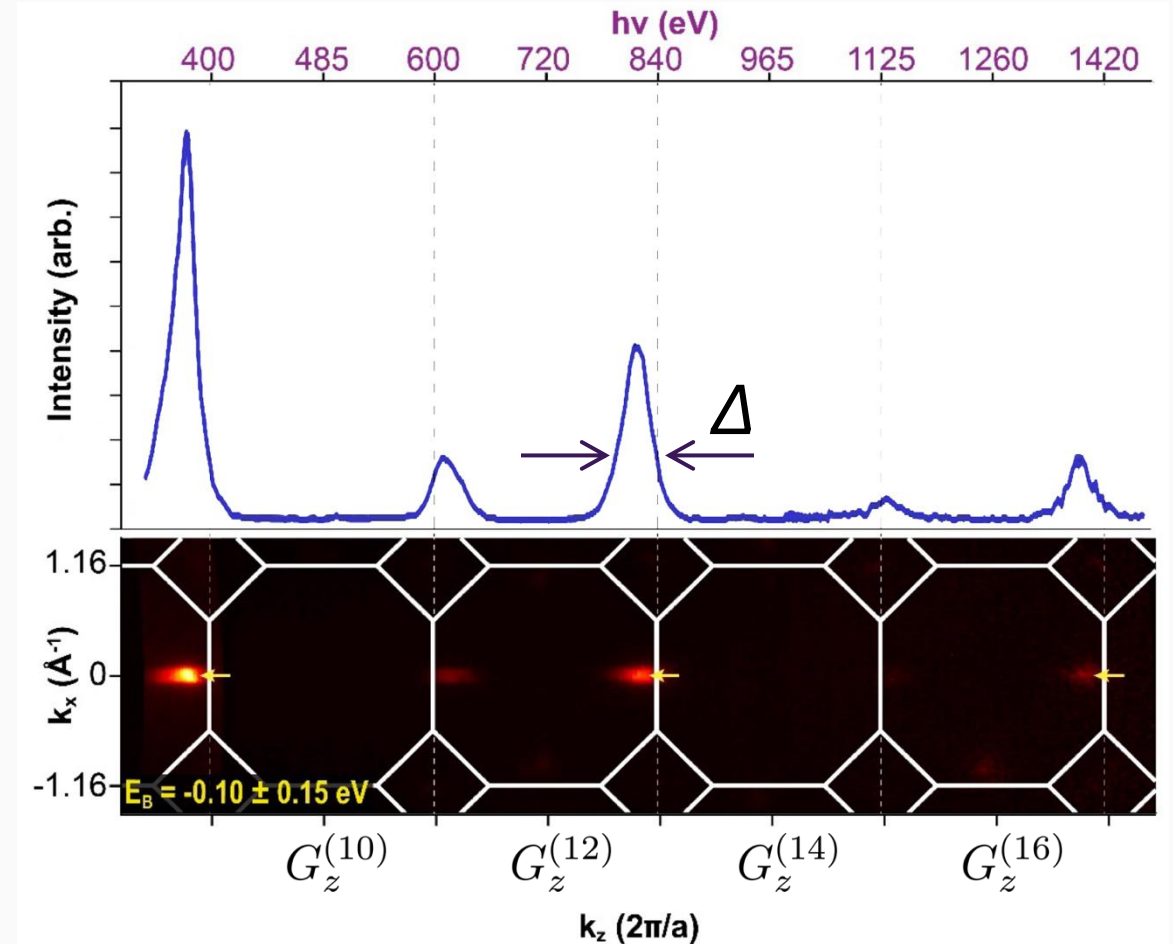


k_z peaks at predicted G_z positions map the Bloch-wave composition

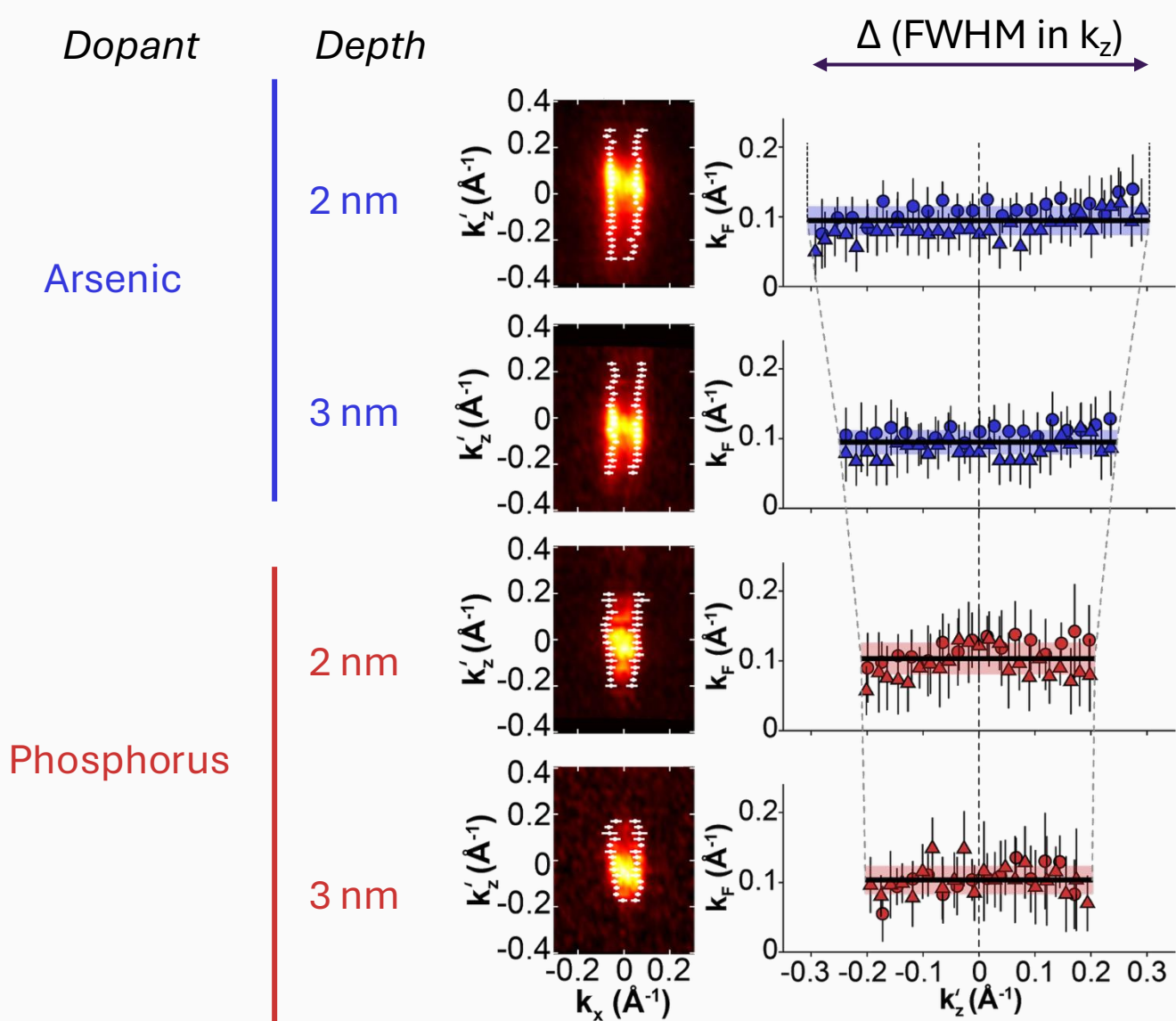
$$I_{\text{PE}}^{\mathbf{K}_{\parallel}} \propto \sum_{G_z} \left| C_{\mathbf{K}_{\parallel}}^{G_z} \right|^2 \left| \mathcal{F}_{K_z - k_z - G_z} \left(e^{-K_z^{\text{im}} z} \mathcal{E}(z) \right) \right|^2$$

- We observe peaks at the predicted $k_z + G_z^{(n)}$ positions
- Destructive interference in next-nearest BZ due to the two-atom Si basis
- Peak FWHM Δ gives confinement thickness D
 $\rightarrow \lambda \text{ (IMFP)} = 1.2 - 2.0 \text{ nm } (h\nu = 380 - 820 \text{ eV})$

$$\Delta = \underbrace{\frac{1}{\lambda}}_{\text{IMFP}} + \underbrace{\frac{1}{D}}_{\text{confinement}}$$



As δ -layers are 2× thinner than P (0.45 vs 0.91 nm)



- Electronic thickness: As < P
- Larger atomic size $\rightarrow$ reduced diffusion during overgrowth

δ -layer species, depth	Electronic thickness SX-ARPES [nm]
As, 2 nm	0.45 $\pm$ 0.04
As, 3 nm	0.62 $\pm$ 0.10
P, 2 nm	0.91 $\pm$ 0.21
P, 3 nm	1.03 $\pm$ 0.35

Expanding the palette: germanium as a donor qubit host

Si:P has established a toolkit. Can Ge offer a materially superior host?

Why germanium over silicon?

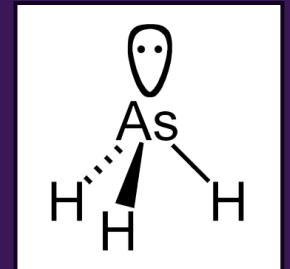
- **Larger Bohr radius (6.4 nm vs 2.5 nm):** relaxed placement tolerance; stronger tunnel and exchange coupling
- **Giant Stark tunability ($\gg$ Si):** all-electrical qubit control, no microwave antenna required
- **Suppressed valley-exchange oscillations:** predictable two-qubit coupling, less valley-interference oscillation

Ge retains key practical advantages:

- **Spin coherence:** $T_2 \sim 1$ ms — sufficient for high-fidelity multi-qubit gates
- **Isotopic purification:** ^{73}Ge depletion removes dominant nuclear-spin decoherence
- **Fabrication:** CMOS-compatible processing

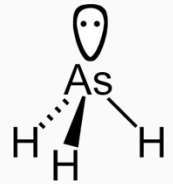
14 Si 28.086	15 P 30.974
32 Ge 72.61	33 As 74.922

Arsenic donors in Ge

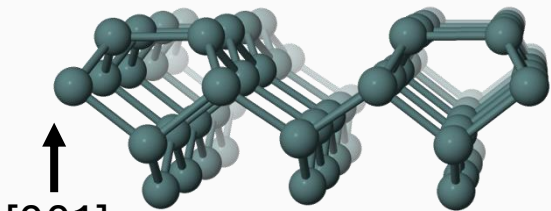


*Ge(001)/AsH₃
chemistry – focus
of our work*

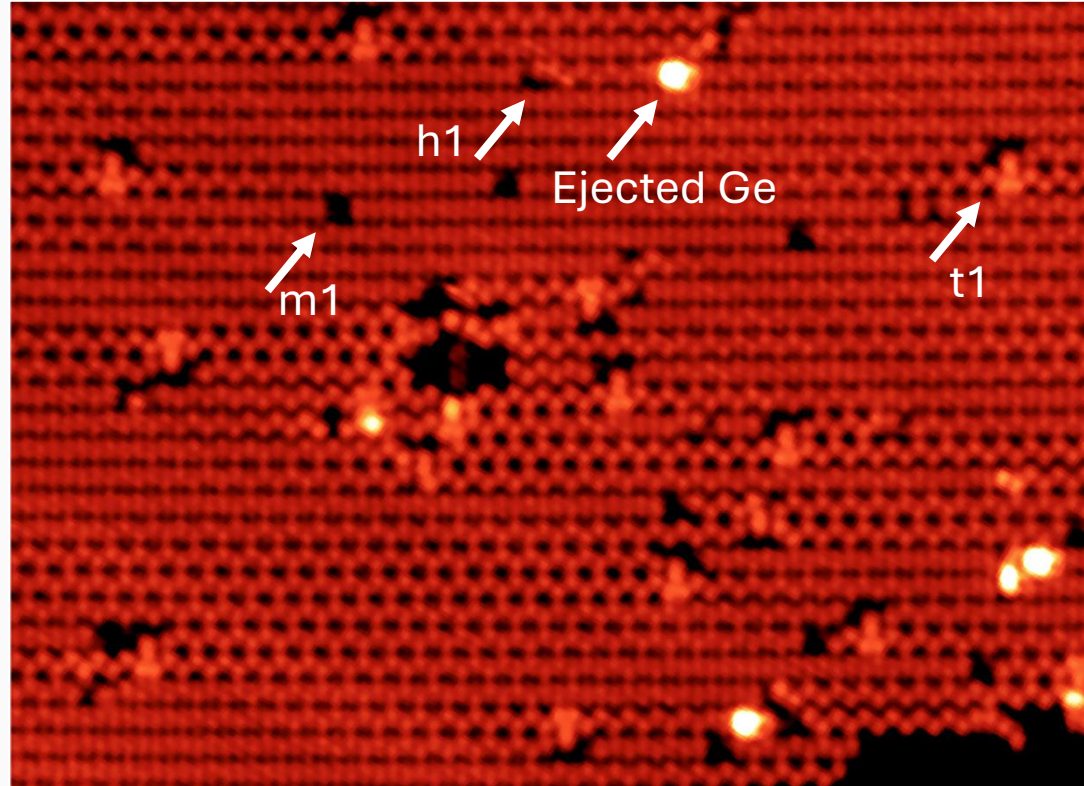
Arsenic incorporates into Ge(001) at room temperature



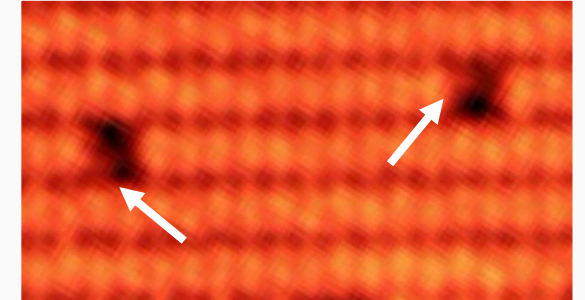
Arsine (AsH_3)



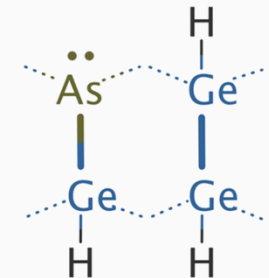
Ge(001) surface



STM image low AsH_3 dose: m1 (most abundant), h1, t1 = incorporated arsenic features



Two m1 incorporated arsenic features



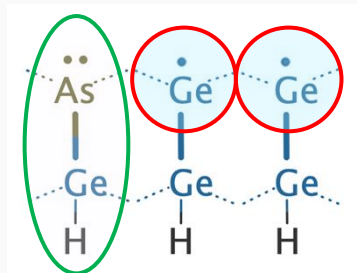
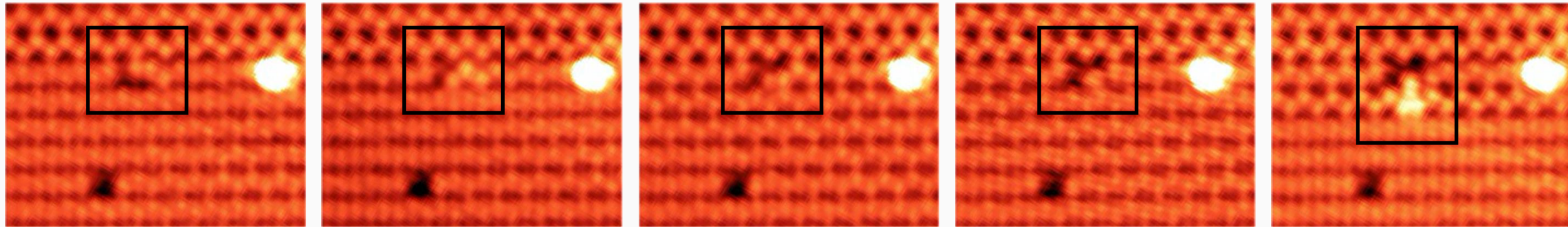
m1 geometry: As-Ge substitution, H retained on adjacent Ge

- RT AsH_3 dissociation drives As incorporation into Ge dimer rows.
- As substituted into Ge dimer row in all three cases; m1, h1, t1 differ by H adsorption site.

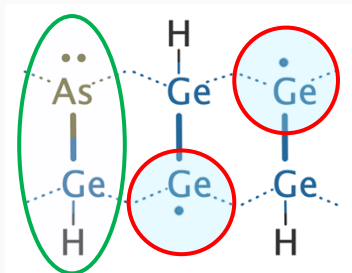
Sequential structural transitions of a single As feature

STM images, same defect site, sequential acquisition

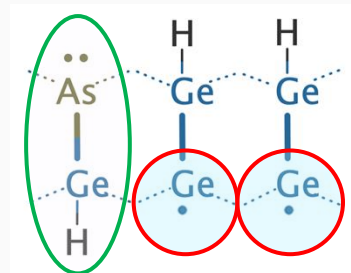
h1 → h2 → h3 → g1 → t1



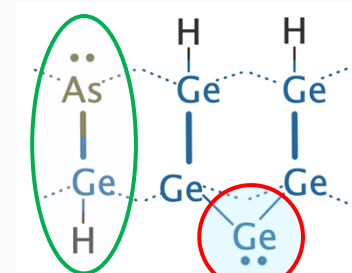
h1 (-2.8 eV)



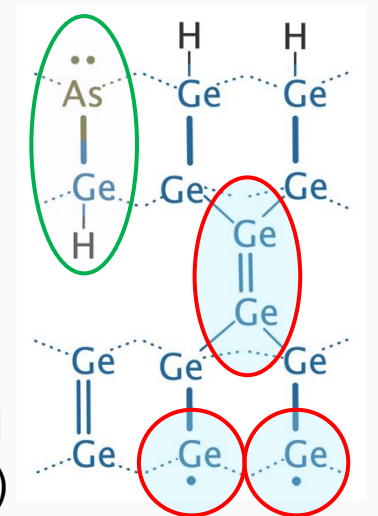
h2 (-2.8 eV)



h3 (-2.8 eV)



g1 (-2.9 eV)



t1

(Ge ad-dimer formation)

 As-Ge-H dimer (depression)

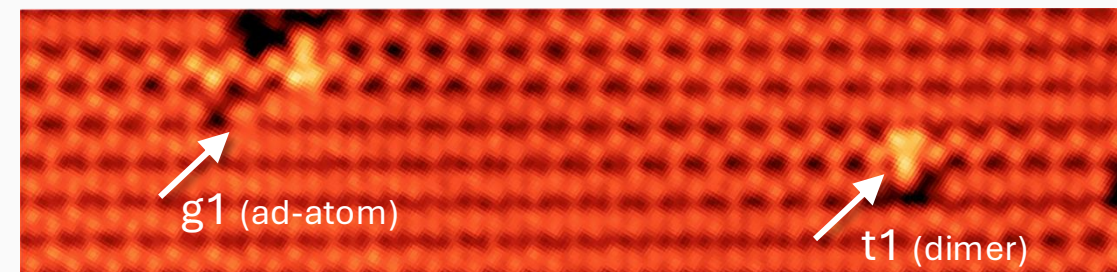
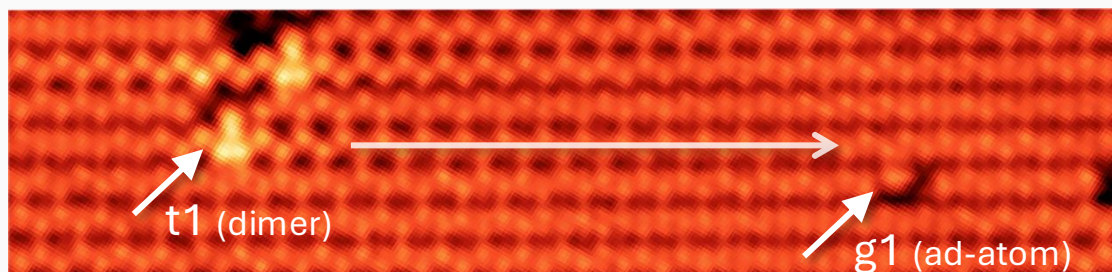
 Ge dangling bond (protrusion)

DFT: B3LYP/6-311G++(2df,2pd)

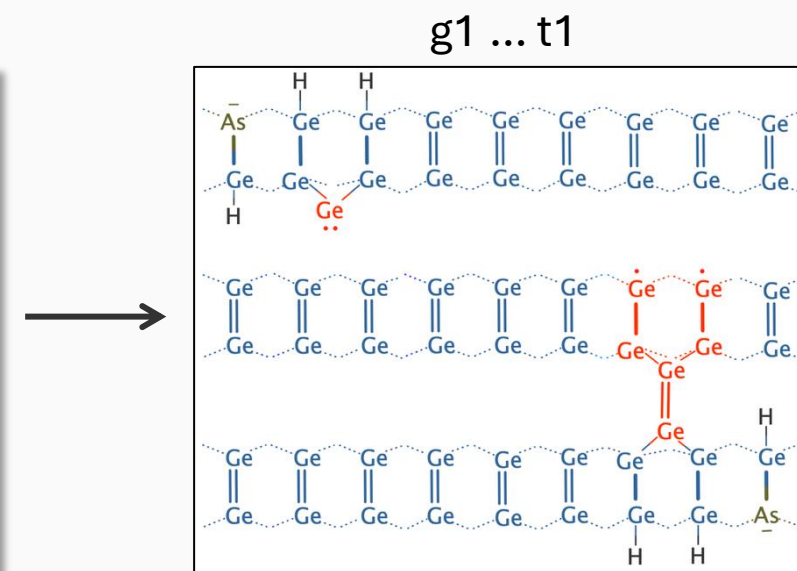
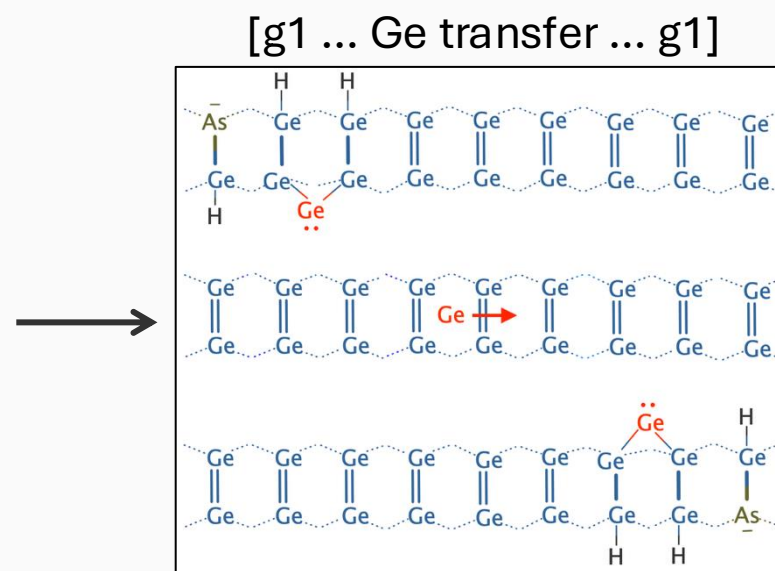
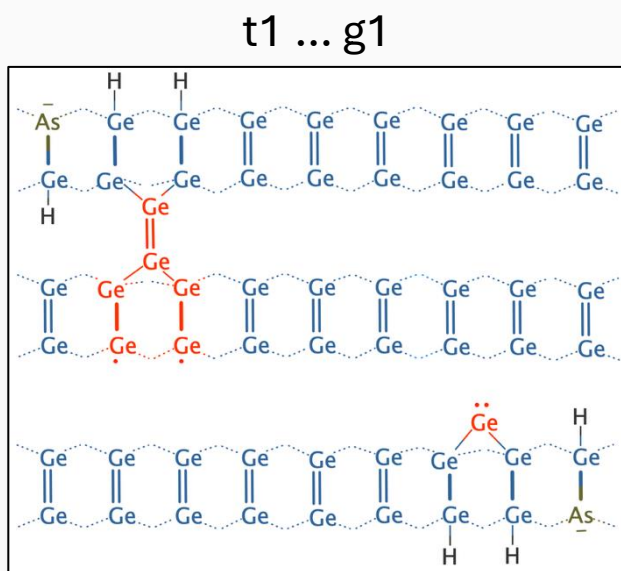
Hofmann, et al., Angewandte Chemie, 62, e202213982 (2023)

Ge ad-atoms are exchanged between t1 and g1 features

Sequential STM images, same area



-1.5 V, 200 pA; $\Delta t = 5$ min



Transition statistics confirm the predicted reaction pathways

Successive STM images over 12 hours; -1.5 V, 200 pA

From:\To:				
	h1/h3	h2/h2*	g1	t1
h1/h3	15	36	8	0
h2/h2*	36	42	1	0
g1	11	1	0	40
t1	0	0	41	0



Hydrogen shift transitions



Ge ad-atom ↔ ad-dimer transitions

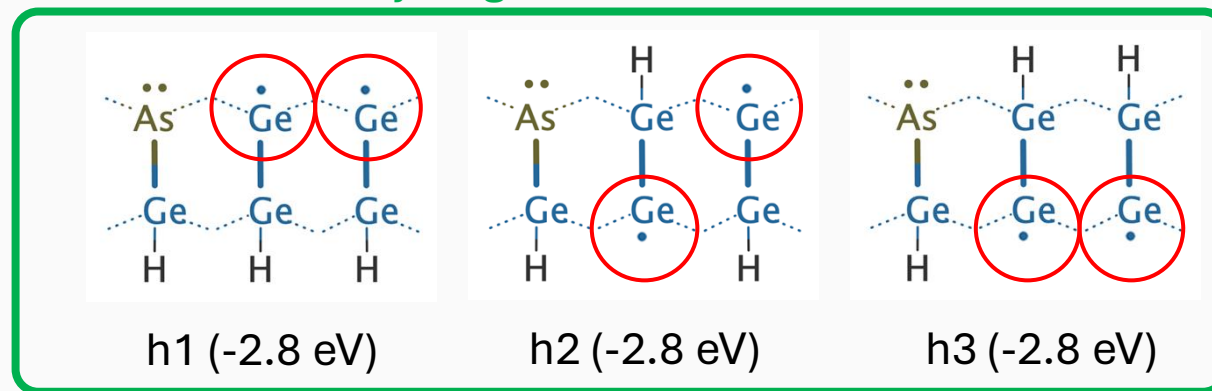


Ge ad-atom shift to hydrogen-dangling bonds

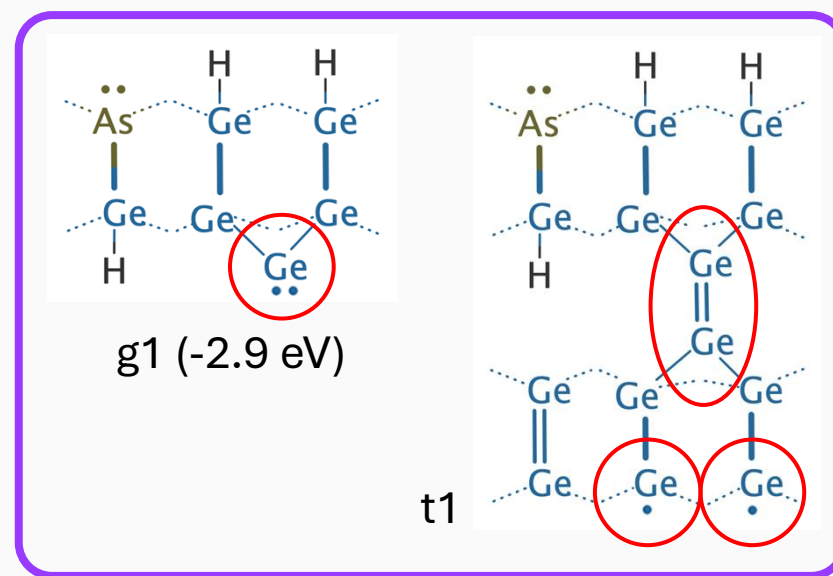


Never observed — multi-step, high barrier at RT

Hydrogen shift transitions



Ge ad-atom/ad-dimer transitions

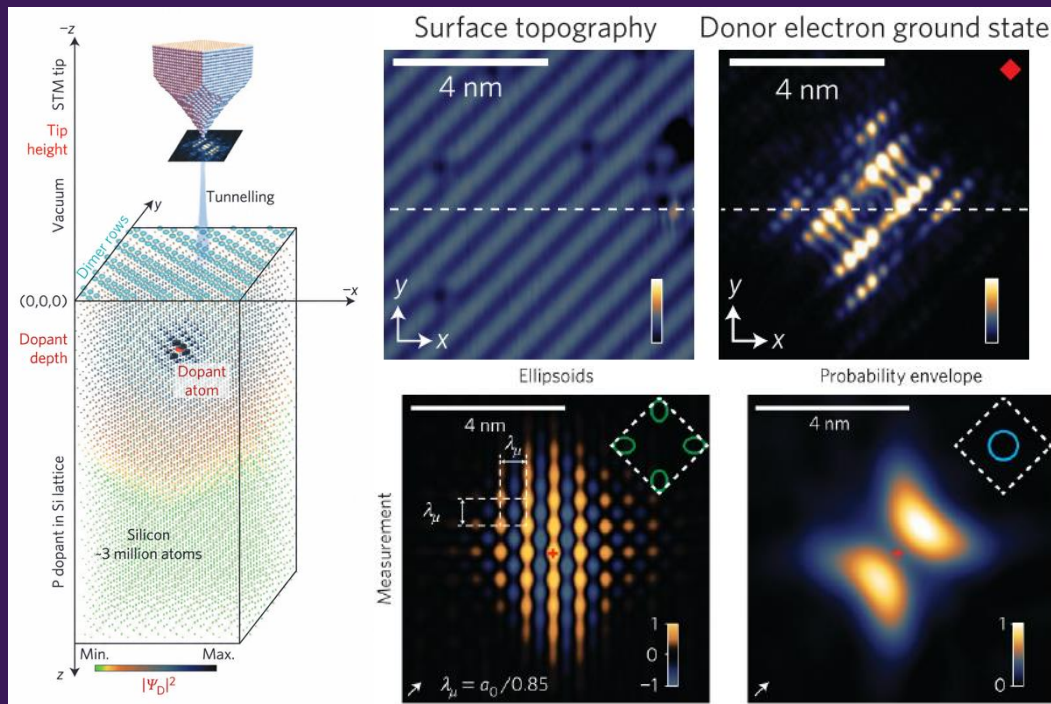


Ge ad-atom transfer

STM images the wave function of a single dopant atom

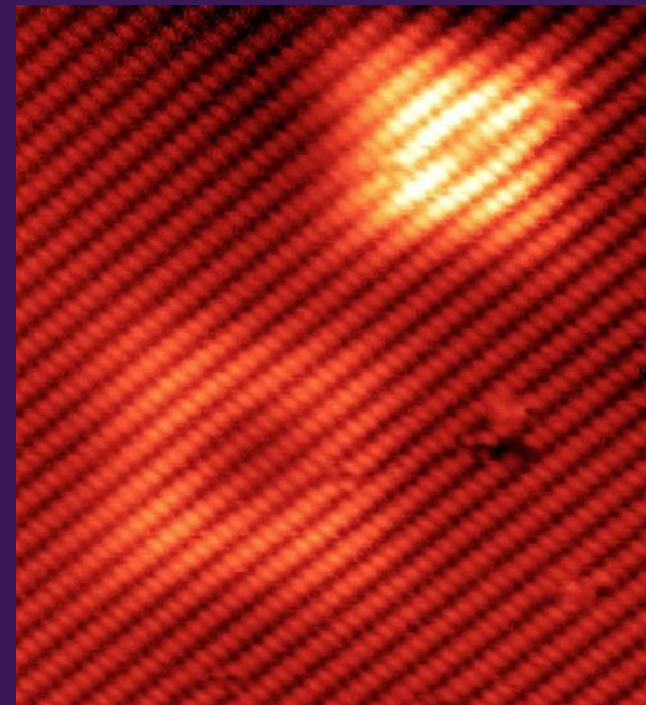
- **A dopant is an artificial hydrogen atom:** it binds one electron or hole in an orbit a few nm across
- **STM maps that orbit atom by atom,** for donors and acceptors alike
- **The images test theory directly:** matching them pins down model approximations and parameters

Hydrogenic donor state in Si



Salfi et al., Nat. Mater. **13**, 605 (2014)

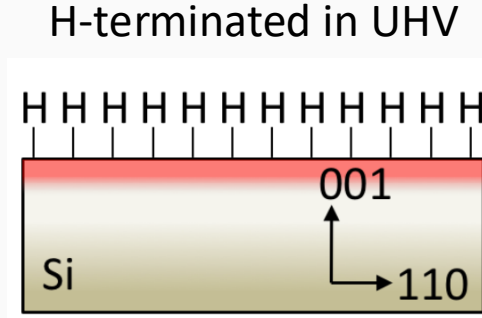
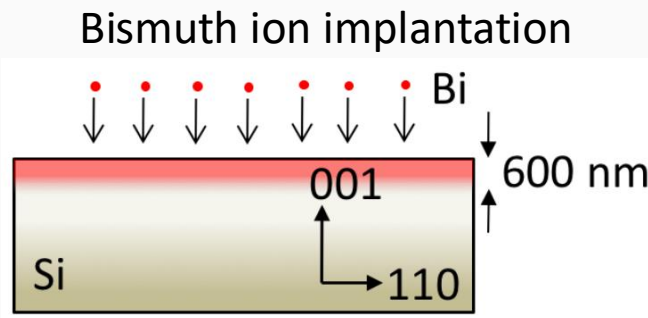
Hydrogenic acceptor state in Si



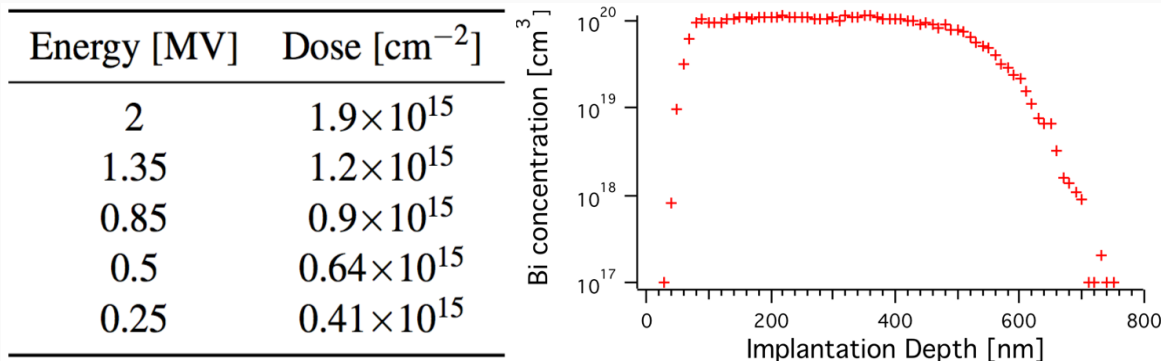
Siegl et al., Nano Lett. **25**, 13996 (2025)

Subsurface acceptor states created via ion implantation of bismuth

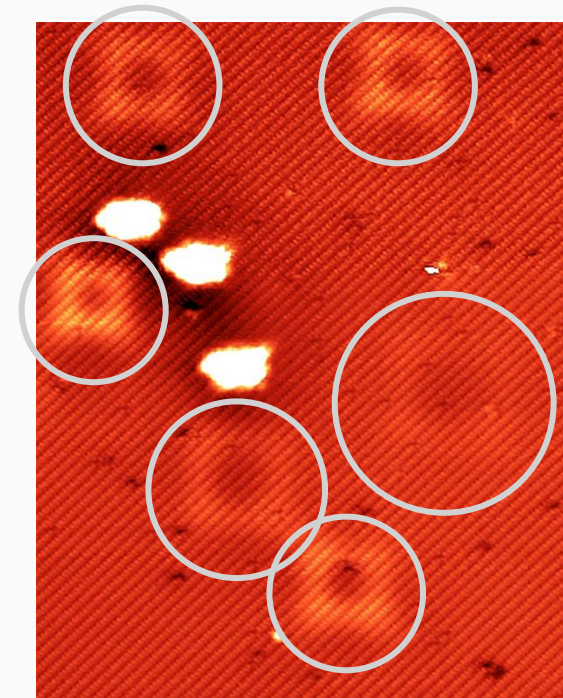
- p-type surface and square-ring features in STM data of Bi implanted Si(001)



Bi ion implantation doses and profile (SRIM)

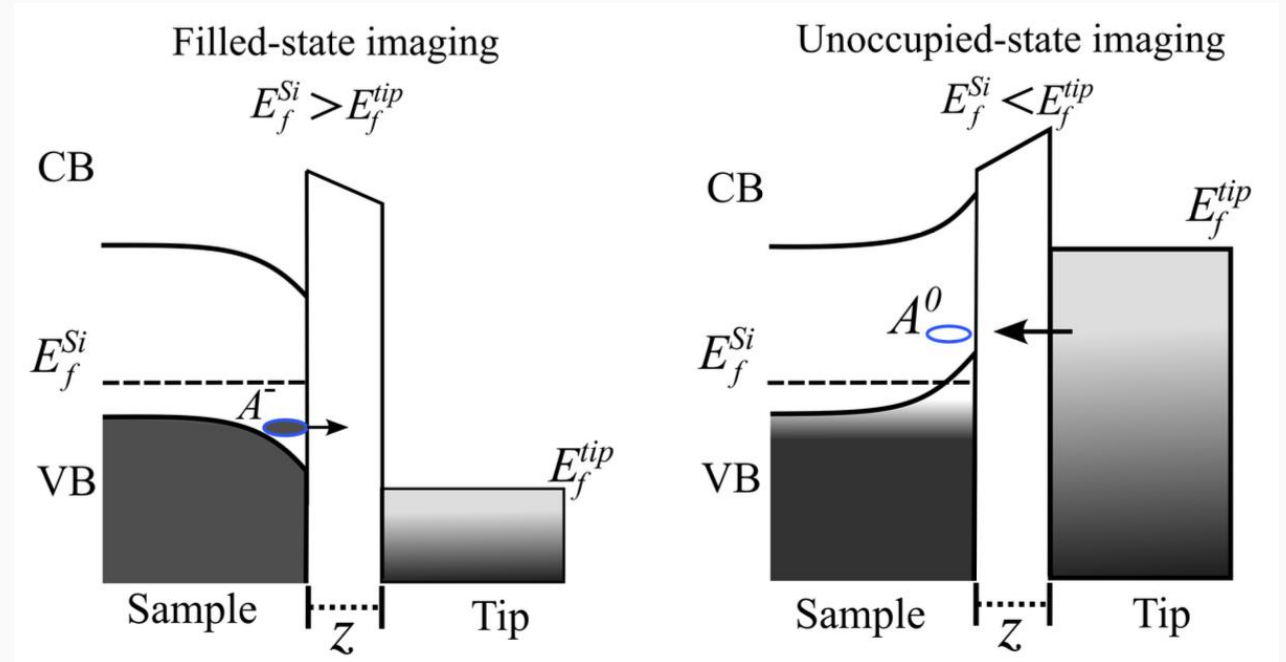
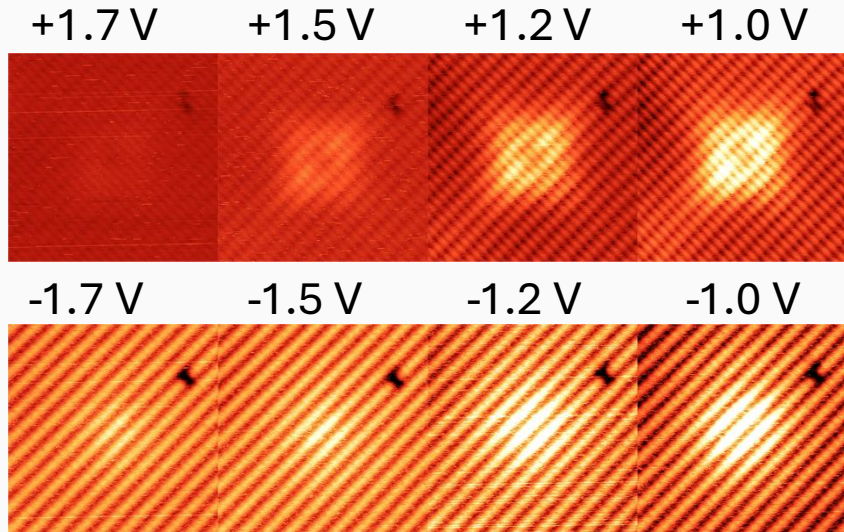


STM image (+1.2 V)



Siegl *et al.*, Nano Lett. **25**, 13996 (2025)

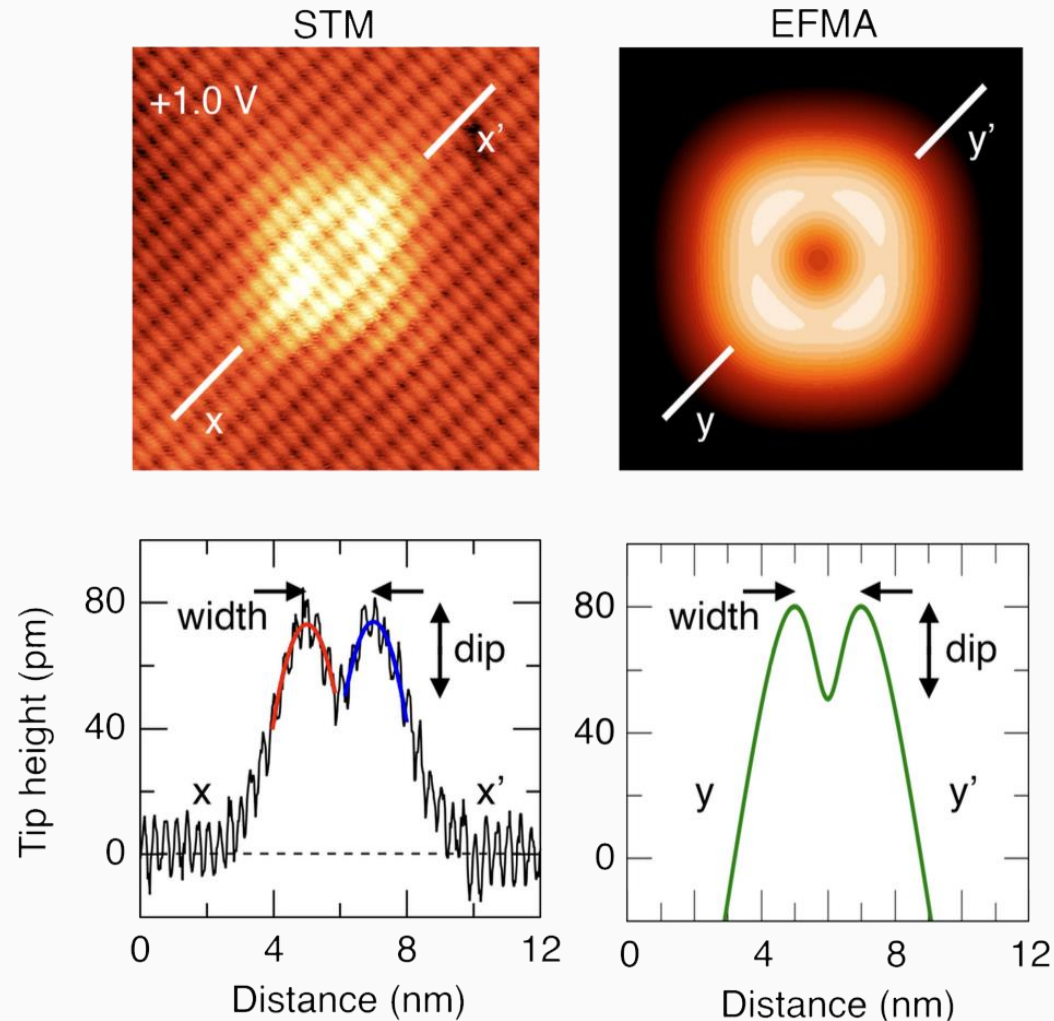
Subsurface acceptor states created via ion implantation of bismuth



Julian Zanon PhD Thesis TU/e 2026

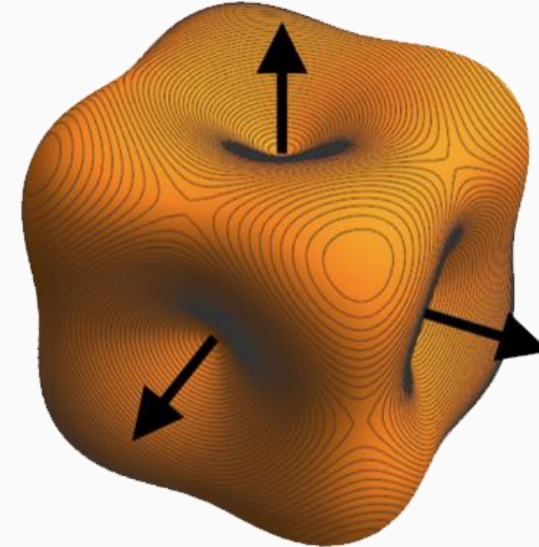
- **Filled states:** acceptor ionised (A^-); sites image as bright protrusions
- **Empty states:** tip-induced band bending neutralises the acceptor (A^0); the same sites image as dark rings

Effective-mass theory reproduces the acceptor image



Siegl *et al.*, Nano Lett. **25**, 13996 (2025)

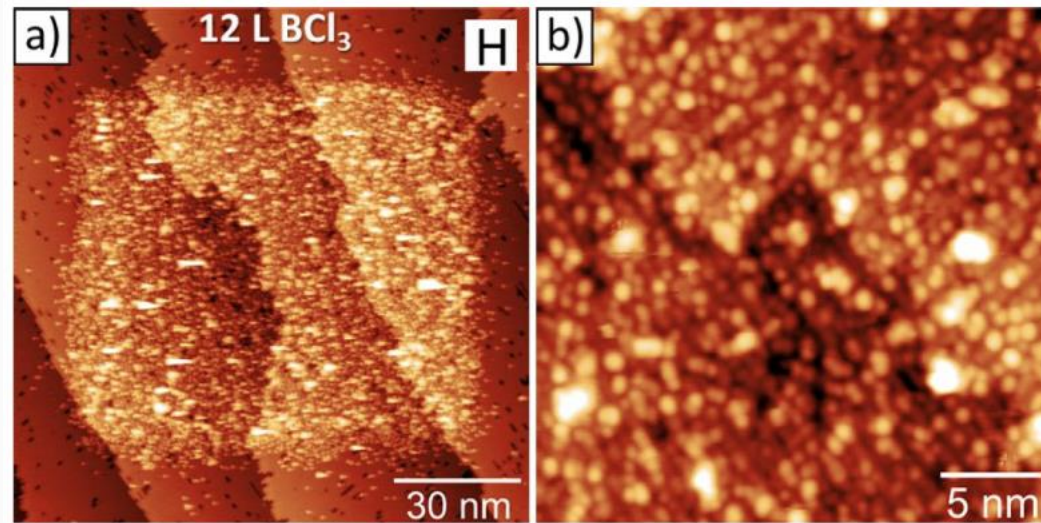
- **STM (+1.0 V):** acceptor images as a square ring with a central dip
- **Effective mass approximation:** model parameters tuned to match the measured profile width and dip
- **3D wavefunction:** anisotropic lobes follow the cubic symmetry of the Si valence band



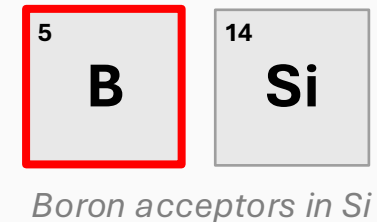
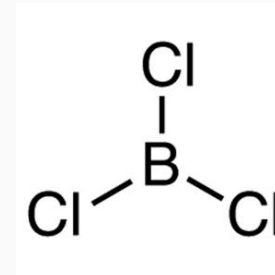
*EFMA acceptor wavefunction;
arrows along $\langle 100 \rangle$*

Next steps for acceptors: deterministic B atom placement via halide precursor

- **Strong spin-orbit coupling ($\Delta_{\text{SO}} = 44 \text{ meV}$):** all-electrical, gate-driven spin control
- **No valley interference:** single Γ -point band maximum, so exchange varies smoothly with placement
- Halide precursors like BCl_3 demonstrated compatibility with STM H-lithography



Dwyer et al., ACS Appl. Mater. Interfaces **13**, 41275 (2021)



Summary

- Kane's 1998 P in Si quantum computer proposal motivated atomic-scale semiconductor devices for quantum technology.
- STM atomic manipulation, developed in the 1990s, provided the tool.
- P in Si devices have reached 11-qubit demonstrations, but fabrication still lacks single-atom positioning at scale.
- New methods and materials are providing new directions, e.g. SX-ARPES characterisation, utilising the Germanium band structure and acceptor physics.

Community roadmap (2025)

IOP Publishing Nano Futures 9 (2025) 012001 <https://doi.org/10.1088/2399-1984/ada901>

NANO FUTURES

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ROADMAP

Roadmap on atomic-scale semiconductor devices

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